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O-GlcNAcylation of DJ-1 suppresses ferroptosis in renal cell carcinoma by affecting the transsulfuration pathway

Zhiwei Yan [a,b,1], Yanze Li [a,b,1], Minghui Wang [a,b,1], Kai Xu [a,b], Yunxun Liu [a,b], Lei Wang [a,b], Hongbo Luo [a,b,*], Zhiyuan Chen [a,b,*], Xiuheng Liu [a,b,*]

[a] Department of Urology, Renmin Hospital of Wuhan University, Wuhan 430060 Hubei, China
[b] Institute of Urologic Disease, Renmin Hospital of Wuhan University, Wuhan 430060 Hubei, China

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ABSTRACT

Renal cell carcinoma (RCC) is one of the most common urological malignancies worldwide, and advanced patients often face challenges with chemotherapy resistance and poor prognosis. Ferroptosis, a novel form of cell death, offers potential therapeutic prospects. In this study, we found that DJ-1 was elevated in kidney renal clear cell carcinoma (KIRC), and this abnormal expression pattern was closely associated with clinical pathological characteristics and worse prognosis. Our experiments both in vivo and in vitro revealed that DJ-1 enhanced the malignant characteristics of KIRC, leading to increased tumor growth. Additionally, DJ-1 inhibited ferroptosis through promoting homocysteine (Hcy) synthesis in the transsulfuration pathway in KIRC cells. Mechanistic studies revealed that O-GlcNAc transferase (OGT) mediated O-GlcNAcylation of DJ-1 was crucial for maintaining its homodimeric structure. Importantly, O-GlcNAcylation-deficient mutation of DJ-1 at T19 residue enhanced the interaction between S-adenosyl homocysteine hydrolase (SAHH) and the negative regulatory factor S-adenosyl homocysteine hydrolase-like-1 (AHCYL1), thereby inhibited the activities of SAHH and transsulfuration pathway. In summary, the oncogenic role of DJ-1 in KIRC was closely related to the reduction of ferroptosis, and the O-GlcNAcylation of DJ-1 exerted an antioxidant effect by activating the transsulfuration pathway. Therefore, DJ-1, specifically O-GlcNAcylation of DJ-1 could represent an important target for ferroptosis-based anti-tumor therapy.

1. Introduction

Renal cell carcinoma (RCC) is a common malignancy of the urinary system, placing 14th in incidence among all tumors and 16th in mortality rates [1]. Kidney renal clear cell carcinoma (KIRC) is the most prevalent and extensively studied pathological subtype of RCC, accounting for approximately 70% of all clinical cases [2]. Currently, issues such as the risk of postoperative recurrence in patients with localized KIRC and chemotherapy resistance in advanced KIRC patients continue to impede the progress of renal cancer treatment [3]. Although immunotherapy and targeted therapy have emerged as new breakthroughs, the efficacy of these treatments varies significantly among individuals [4]. To advance the creation of new therapeutic and diagnostic approaches, it is essential to deepen our understanding of the mechanisms driving renal cancer.

Ferroptosis represents a distinct type of cell death resulting from the buildup of iron-dependent reactive oxygen species (ROS), which disrupts the cellular redox balance. This form of cell death is distinguished by the accumulation of lipid peroxides and changes in mitochondrial structure, including a reduced volume, outer membrane rupture, and fewer mitochondrial cristae. Additionally, the nucleus remains normal in size and lacks chromatin condensation, setting ferroptosis apart from

Abbreviations: *RCC*, Renal cell carcinoma; *KIRC*, Kidney renal clear cell carcinoma; *Hcy*, Homocysteine; *O-GlcNAc*, O-linked β-N-acetylglucosamine; *OGT*, O-GlcNAc transferase; *SAHH*, S-adenosyl homocysteine hydrolase; *AHCYL1*, S-adenosyl homocysteine hydrolase-like-1; *ROS*, reactive oxygen species; *TCGA*, The Cancer Genome Atlas database; *GEO*, Gene Expression Omnibus; *MDA*, Malondialdehyde; *Fer-1*, Ferrostatin-1; *3-MA*, 3-Methyladenine; *Z-VAD-FMK*, Carbobenzoxy-Valyl-Alanyl-Aspartyl-(O-Methyl)-Fluoromethylketone; *Nec-1*, Necrostatin-1; *NAC*, N-acetylcysteine; *Met*, Methionine; *SAM*, S-adenosyl methionine; *SAH*, S-adenosyl-L-homocysteine.

* Corresponding authors.
E-mail addresses: luohongbo@whu.edu.cn (H. Luo), drchenzy@whu.edu.cn (Z. Chen), drliuxh@whu.edu.cn (X. Liu).
[1] These authors contributed equally to this work.

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other cell death mechanisms. [5]. Notably, cancer cells that exhibit drug resistance and high metastatic potential can evade multiple forms of cell death but display a heightened sensitivity to ferroptosis [6]. Various ferroptosis inducers have shown significant potential in cancer treatment. Nevertheless, precisely regulating ferroptosis to intervene in the progression of related diseases remains a critical area for further exploration. In-depth research into the molecular mechanisms underlying ferroptosis will contribute to achieving this goal.

DJ-1 was first identified as an oncogene that amplifies the Ras/MAPK signaling pathway. It was subsequently recognized as a pathogenic gene linked to autosomal recessive early-onset Parkinson's disease, which has led to significant research advances in neurology [7]. The DJ-1 protein comprises 189 amino acids and predominantly exists in homodimers form, exhibiting widespread expression across various human tissues [8]. This protein features three cysteine residues: Cys46, Cys53, and Cys106, with Cys106 playing a pivotal role in DJ-1's antioxidant functions [9]. Under conditions of oxidative stress, cytoplasmic DJ-1 is capable of translocating to the mitochondria, thereby preventing cell death induced by oxidative stress and maintaining mitochondrial homeostasis [10]. Although the antioxidant properties of DJ-1 are well-established, it remains uncertain whether its antioxidant capacity can counteract ferroptosis triggered by the accumulation of lipid ROS. Consequently, exploring the relationship between DJ-1 and ferroptosis holds significant potential for unlocking new therapeutic strategies in cancer treatment.

In this study, we identified that DJ-1 exhibited high expression levels in KIRC. Both in vitro and in vivo research confirmed that DJ-1 markedly aggravated the malignant characteristics of KIRC, demonstrating a strong correlation with ferroptosis activity. Rescue experiments and metabolic analyses revealed that DJ-1 primarily affected homocysteine (Hcy) production in the transsulfuration pathway, thereby regulating ferroptosis. Notably, further mechanistic studies revealed that O-GlcNAc transferase (OGT) mediated O-GlcNAcylation modification enhanced the stability of DJ-1's homodimeric structure, which was critical for its antioxidant function. Moreover, O-GlcNAcylation-deficient mutation of DJ-1 at T19 residue increased the interaction between SAHH and its negative regulator AHCYL1, thus inhibiting the activity of SAHH and Hcy production, and meanwhile, the level of ferroptosis was also elevated in the O-GlcNAcylation-deficient mutation cells. Consequently, O-GlcNAcylation of DJ-1 acts as an inhibitor of ferroptosis, providing crucial insights into the development of innovative diagnostic and treatment approaches for KIRC.

2. Methods

2.1. Analysis of public databases

All clinical and RNA-seq data were sourced from The Cancer Genome Atlas (TCGA) database and Gene Expression Omnibus (GEO) database. To assess the differential expression of DJ-1 in paired samples of KIRC, the expression data of the tumor underwent Log2 transformation and T-test analysis. Subsequently, the differences between DJ-1 expression and pathological features were evaluated, and Kaplan-Meier (K-M) survival curves were plotted.

2.2. Immunohistochemistry analysis

The paraffin-embedded sections were placed at 65 °C for 40 min followed by deparaffinization and hydration. To thoroughly remove organic solvents, the sections underwent repeated washes with PBS. Subsequently, antigen retrieval at high temperature was carried out using citrate buffer, followed by another round of washing after cooling. To inhibit endogenous peroxidase activity, 3% H2O2 was applied for 10 min. This was followed by a 1-hour blocking step with 10% goat serum. The following antibodies were employed for the incubation process: anti-DJ-1 (Affinity, China, DF6169, diluted at 1: 200) and anti-Ki67 (Proteintech, China, 27309–1-AP, diluted at 1: 2000).

2.3. Cell culture and transfection

The cells used in this experiment, all STR-identified, were obtained from ATCC. The cells were maintained in distinct culture conditions: HK-2 cells were propagated in DMEM medium (HyClone, China), whereas RPMI 1640 medium (HyClone, China) was used to maintain the other KIRC cell lines (786-O,769-P and Caki-1). To establish DJ-1 knockdown and overexpression cell lines, two specific siRNA sequences targeting DJ-1, along with pcDNA3.1-Flag-DJ-1, pcDNA3.1-Myc-DJ-1 plasmid, were procured from Sangon Biotech (Shanghai, China). Additionally, we also constructed two specific siRNA sequences targeting OGT and cloned O-GlcNAcylation-deficient mutation of DJ-1 (Thr19 → Ala: T19A) into pcDNA3.1-Flag vector. The complete sequences of all siRNA were detailed in the Supplementary Table S1. When the cell confluence reached approximately 70%, Lipofectamine 8000 with siRNA or Lipofectamine 3000 with plasmids were added into indicated cells according to the manufacturer's instructions, following which the cells were further incubated for 48 h.

2.4. Western blot

The concentration of the extracted proteins was then assessed by a BCA Protein Assay Kit (Beyotime, Shanghai, China). Following SDS-PAGE separation, the proteins were transferred to PVDF membranes (Millipore, USA), which were blocked for 1 h by 5% non-fat milk and incubated overnight at 4 °C by primary antibodies. The protein bands were subsequently quantified by the ChemiDoc™ XRS + System and Image J software. The primary antibodies employed in this study included: anti-DJ-1 (Proteintech, China, 11681–1-AP), anti-GPX4 (HUABIO, China, ET1706-45), anti-SLC7A11 (Proteintech, China, 26864–1-AP), and anti-GAPDH (Proteintech, China, 10494–1-AP), anti-Beta Tubulin (Proteintech, China, 10094–1-AP), anti-FLAG (Proteintech, China, 66008–4-Ig), anti-MYC (Proteintech, China, 60003–2-Ig), anti-OGT (Proteintech, China, 11576–2-AP), anti-O-GlcNAc (Abcam, USA, ab2739), anti-AHCYL1 (Cell Signaling Technology, USA, #94248), anti-SAHH (Santa Cruz Biotechnology, USA, sc271389).

2.5. Quantitative real-time PCR (qRT-PCR)

Under RNase-free conditions, cells total RNA was isolated by TRIzol reagent (Invitrogen, USA). Reverse transcription of RNA into cDNA was implemented employing the PrimeScriptTM RT Kit (TaKaRa, Japan). Subsequently, qRT-PCR was conducted on an A100 PCR system. The primer sequences were listed in Supplementary Table S2.

2.6. Cell proliferation assay

CCK-8 assays (Beyotime, Shanghai, China) were conducted using a 96-well plate. Each well contained 5,000 cells that had been treated accordingly. After the designated incubation time, CCK-8 solution was added, and absorbance values were measured. The EdU assay (Beyotime, Shanghai, China) was employed to further investigate cell proliferation. After performing EdU and Hoechst 33,342 staining in the 24-well plate, observations and imaging were conducted.

2.7. Cell migration and invasion assays

To evaluate cellular migration capabilities, a transwell assay was performed. Cell lines with knockdown and overexpression were plated in six-well dishes and grown until they approached confluence. After counting the cells, a suitable number of cells were transferred to the upper chamber containing serum-free medium for cultivation. The lower chamber was filled with medium containing a high concentration of serum to induce cell migration. After 24 h, the cells were fixed and

stained, and then observed using a microscope. Next, Matrigel (Corning, USA) was first diluted with serum-containing medium, then polymerized and hydrated to ensure that no liquid leaked into the lower chamber. The subsequent experimental procedure was identical to that of the migration assay.

2.8. Cell activity and toxicity assays

Cell viability and toxicity were evaluated using the Calcein/PI assay kit (Beyotime, Shanghai, China). After the cells in the six-well plate reached 80% confluence, different treatments were applied. Then, 1 ml of Calcein/PI working solution was introduced for incubation in the dark. The final results were observed and recorded using a fluorescence microscope.

2.9. Transmission electron microscopy

Cells were seeded in 10 cm culture dishes. After treatment, the cells were digested with trypsin to obtain a cell pellet. Subsequently, a 2.5% glutaraldehyde solution was added for fixation at 4 °C for 4 h. The samples were then dehydrated using various concentrations of ethanol, with each concentration applied for 15 min. After dehydration, the cell samples were embedded and sectioned into ultrathin slices. Staining was performed using a 2% uranyl acetate solution and lead citrate, followed by observation of mitochondrial morphological changes using a transmission electron microscope.

2.10. Malondialdehyde (MDA) level assay

The MDA content in the cells was assessed using the MDA detection kit (Servicebio, Wuhan, China). The treated cells were harvested, and on ice, an appropriate volume of IP lysis buffer was added and incubated for 30 min. The working solution for MDA detection was applied to both the standard and experimental samples, and mixed thoroughly. Subsequently, the samples were rapidly cooled in an ice bath for 5 min. After centrifugation, the supernatant was collected for analysis.

2.11. Lipid ROS assay

Cells in the six-well plates were processed and digested with trypsin to collect the cell pellet. After resuspending in culture medium, the cell count was determined. After centrifugation to remove the supernatant, cells were resuspended in an appropriate volume of C11-BODIPY working solution (Beyotime, Shanghai, China) to achieve a density of 1×10^6 cells/ml. After placing in the incubator for 20 min, placed the cells in PBS for flow cytometry analysis.

2.12. Iron assay

The iron content in cells was assessed using the total iron detection kit (Servicebio, Wuhan, China), following the provided instructions. Briefly, after cell collection, the cells were lysed with a buffer, and the resulting supernatant was used for analysis. To both 100 μl of the standard and the sample, 100 μl of buffer was added. Subsequently, each sample received the addition of 100 μl of detection probe, followed by a 40-minute incubation before centrifugation. After centrifugation, the supernatant was collected for analysis.

2.13. GSH assay

To quantify GSH levels in the cells, the GSH detection kit (Servicebio, Wuhan, China) was employed. In brief, GSH reacts with the chromogenic substrate DTNB to produce TNB and GSSG, with TNB being a yellow product. By measuring the amount of TNB, the GSH content in the sample can be determined. Therefore, the GSH concentration was calculated based on the measured absorbance.

2.14. Hcy and SAH assay

After processing the cells, centrifuge them for 20 min to collect the cell pellets. Perform repeated freeze–thaw cycles to disrupt the cells and release cellular components. Centrifuge again at 4 °C for 20 min to collect the supernatant for subsequent use. ELISA kits (Jiancheng, Nanjing, China) was used to measure and calculate the levels of Hcy and SAH, as the final step.

2.15. Co-Immunoprecipitation (Co-IP)

Co-IP assay was executed by Classic Magnetic Protein A/G IP/Co-IP Kit (Epizyme, YJ201) according to instruction. Firstly, Cells were treated with lysis/wash buffer containing protease/phosphatase inhibitor cocktail (Roche, 4693159001) and UniversalBenzo nuclease (Vazyme, RM1022-00). The protein supernatant obtained after centrifugation was subjected to determination of protein concentration. Meanwhile, 5 μl whole lysate was retained as input.

For immunoprecipitation, the whole lysate was rotated overnight at 4 °C with the indicated primary antibodies and pretreated protein A/G magnetic beads. The antigen–antibody-magnetic bead complex were captured after standing on the magnetic frame for 1 min. Finally, he proteins that were precipitated were ultimately collected for further analysis. In endogenous immunoprecipitation experiments, IgG (CST, 2729) was served as a negative control.

2.16. Click-IT O-GlcNAc enzymatic labeling detection

The O-GlcNAcylation of DJ-1 was detected and analyzed according to the previous researches [11]. Briefly, the lysates were incubated with Click-iT O-GlcNAc enzymatic labeling system overnight at 4 °C, which labeled the O-GlcNAc modified proteins with azido-modified galactose (GalNAz). Following this, the labeled proteins were reacted with alkyne-biotin reagents as described in the protocol of Click-iT Protein Analysis Detection Kits (Thermo Fisher Scientific, C33372). Then, precipitated by the streptavidin resin, the biotinylated proteins were enriched and eluted by 1 × SDS-PAGE loading buffer containing 20 mM biotin at 95 °C for 10 min. The eluted samples were ultimately analyzed using immunoblotting.

2.17. Establishment of in vivo tumor xenograft model

Fifteen nude mice were randomly divided into three groups: normal, knockdown, and overexpression groups. All mice were ensured to be 6 weeks old. Collected the cells from the 10 cm dishes after specific treatments, and resuspended in PBS to achieve a cell density of 5×10^7/ml, then placed on ice for later use. The cell suspension was well mixed using a 1 ml syringe. Then, 150 μl of this mixture was injected slowly into the axillary regions of the nude mice. Tumor volume measurements were taken and documented at five-day intervals. After 30 days, the tumors were excised from the mice and photographed by group. Tumor weight and volume were measured, with volume calculated using the formula: $V = (L \times W^2) / 2$.

2.18. Statistical analysis

Data analysis and visualization were performed using GraphPad Prism 9.0 and R version 4.1.0. For comparisons between two groups, either an unpaired or paired Student's *t*-test was used. Differences among multiple groups were evaluated using one-way ANOVA followed by Tukey's post hoc test. The $\chi 2$ test was employed to examine the relationship between clinical pathological features of KIRC patients and DJ-1 expression. Survival probabilities were calculated with the Kaplan-Meier method to assess the gene's prognostic value.

3. Results

3.1. DJ-1 exhibited elevated expression levels in KIRC and correlated with poor prognosis and clinical characteristics

To investigate the expression profile of DJ-1 in tumors and its clinical relevance, we initially analyzed DJ-1 expression patterns in matched specimens of KIRC using the TCGA and GEO public databases. The results consistently indicated that DJ-1 was significantly elevated in KIRC compared to normal samples (Fig. 1A). To further investigate the clinical significance of DJ-1, the TCGA cohort was categorized into two groups based on DJ-1 expression (Supplementary Table S3). Clinical correlation analysis revealed significant differences in survival status, grade, and stage between two groups (Fig. 1B). Next, we compared the overall survival of KIRC patients between the high- and low-DJ-1 groups. The findings revealed that the patients in the high-DJ-1 group had worse prognoses. (Fig. 1C). Furthermore, we corroborated the findings in various KIRC cell lines, where the results were highly consistent with the

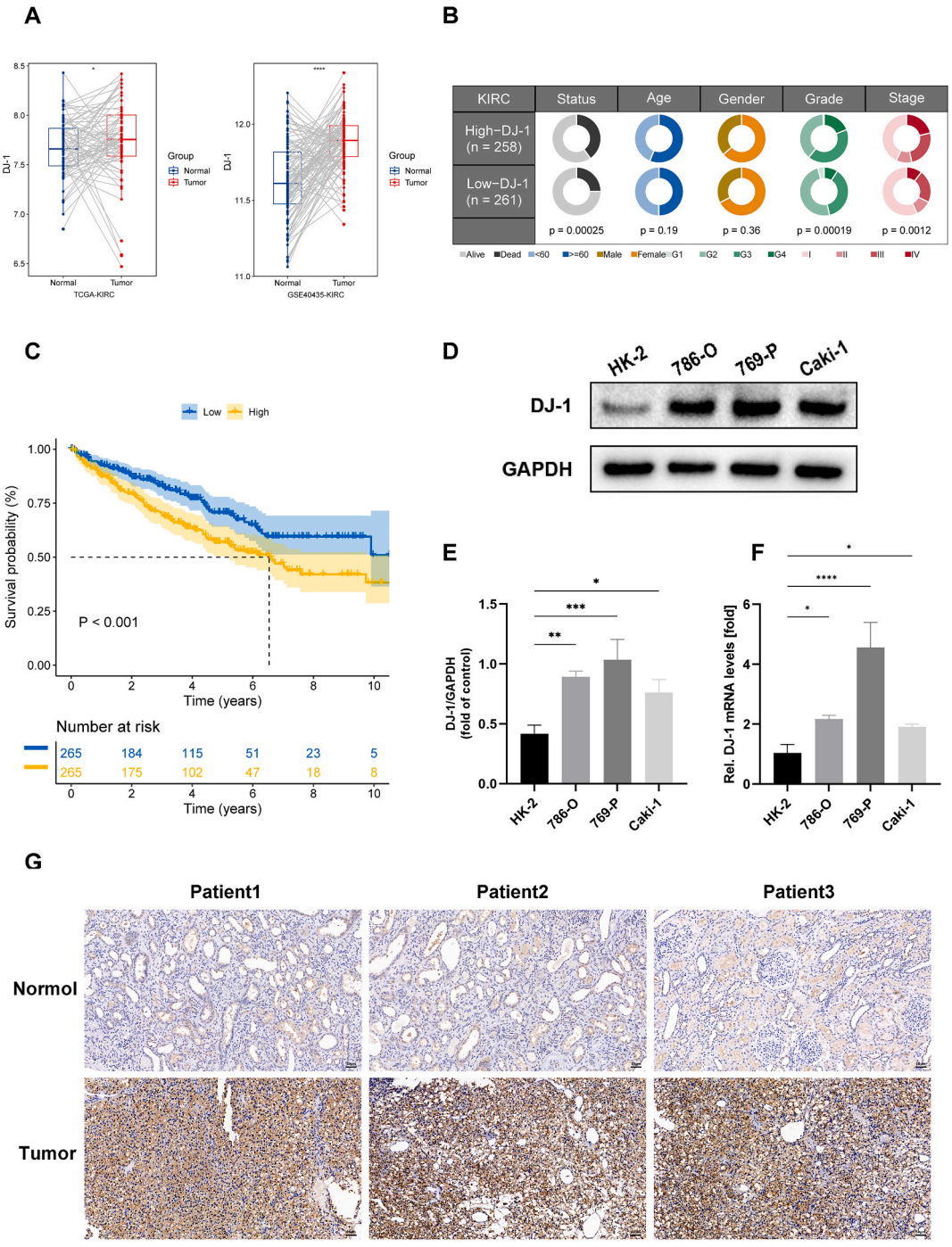


Fig. 1. DJ-1 is significantly overexpressed in KIRC and correlates with clinical pathological characteristics and unfavorable prognosis. A Expression levels of DJ-1 in paired samples of KIRC from the TCGA and GEO databases. B Differences in clinicopathologic characteristics among high- and low-risk groups. C Survival analysis based on DJ-1 expression levels. D, E Quantitative analysis of DJ-1 protein expression. F mRNA expression levels of DJ-1 in KIRC cell lines and HK-2 cells. G Expression of DJ-1 in three pairs of human tumor samples. Scale bar: 50 μm. (All data are presented as mean ± standard deviation of three independent experiments. *P < 0.05, **P < 0.01, ***P < 0.001, ****P < 0.0001).

bioinformatics analysis. Compared to renal tubular epithelial cells (HK-2), DJ-1 mRNA and protein levels were elevated in all three KIRC cell lines examined, with the most pronounced overexpression observed in the 786-O and 769-P cell lines (Fig. 1D–F). Consequently, subsequent investigation focused on these two cell lines. Additionally, immunohistochemical analysis of human tumor tissue samples revealed significantly increased DJ-1 expression in KIRC tissues compared to matched adjacent tissues (Fig. 1G). In conclusion, our results confirmed the aberrant expression pattern of DJ-1 in KIRC and highlighted its strong association with clinical pathological features and poor prognosis, while also providing a solid rationale for studying the oncogenic mechanisms of DJ-1 in KIRC.

3.2. Downregulation of DJ-1 suppressed the malignant characteristics of KIRC cells

To explore the effect of DJ-1 in the malignant phenotype of KIRC, we designed two siRNA sequences specifically targeting DJ-1 and transfected them into two KIRC cell lines, 786-O and 769-P. Firstly, we validated the knockdown efficiency of these two siRNA in mRNA and protein levels, and obtained favorable results (Fig. 2A and B). After knockdown of DJ-1, both EdU assays and CCK-8 results revealed a marked decrease in the proliferation of KIRC cells (Fig. 2C–H). Moreover, in comparison to the control group, KIRC cells with DJ-1 knockdown also displayed a marked reduction in their migration and invasion abilities (Fig. 2I–P). The consistency of these findings across both knockdown groups effectively ruled out the possibility of off-target effects. Collectively, these results demonstrated that downregulating DJ-1 expression inhibited the malignant phenotype of KIRC cells.

3.3. Upregulation of DJ-1 enhanced the malignant characteristics of KIRC cells

Apart from downregulating DJ-1 expression, we explored the effects of DJ-1 overexpression on the malignant phenotype of KIRC. Both qRT-PCR and Western blot analyses confirmed the successful establishment of a DJ-1-overexpressing cellular model in the indicated cell lines (Fig. 3A and B). After DJ-1 overexpression, a pronounced increase in tumor cell proliferation was evident, as demonstrated by EdU assays and CCK-8 results (Fig. 3C–H). Furthermore, the wound healing assay and transwell assay consistently demonstrated increased migratory and invasive capabilities in both two KIRC cell lines (Fig. 3I–P). These observations strongly supported our hypothesis that the upregulation of DJ-1 fostered the malignant characteristics of KIRC cells.

3.4. Downregulation of DJ-1 stimulated ferroptosis activity in KIRC cells

DJ-1 has garnered considerable attention across various fields due to its antioxidant properties, and there are reports indicating that DJ-1 can modulate ferroptosis [12]. Thus, we explored the relationship between DJ-1 expression and ferroptosis in KIRC cells. Given its superior knockdown efficiency, only the second siRNA was employed in the subsequent experiments. To determine whether DJ-1 is implicated in the induction of ferroptosis in KIRC, four common cell death inhibitors were selected to treat the KIRC cell lines following DJ-1 knockdown, after which we assessed cell viability. The results demonstrated that Ferrostatin-1, a classical ferroptosis inhibitor, could prevent cell death caused by DJ-1 knockdown. However, it was not reversed by autophagy inhibitors (3-MA), apoptosis inhibitors (Z-VAD-FMK), or necrosis inhibitors (Nec-1) (Fig. 4A and B). Next, transmission electron microscopy revealed significant morphological alterations (reduced mitochondrial volume, increased membrane density, cristae condensation, and outer membrane rupture) associated with ferroptosis in KIRC cells after DJ-1 knockdown (Fig. 4C). Subsequently, changes in the levels of the typical lipid peroxidation product MDA, lipid ROS and iron content in the knockdown cell lines were examined. As illustrated (Fig. 4D–I), DJ-1 downregulation resulted in a marked increase in the accumulation of MDA, lipid ROS, and iron content in KIRC cells, whereas treatment with Ferrostatin-1 significantly reduced these ferroptosis indicators. Additionally, we analyzed the variations of the ferroptosis-protective GSH. In the DJ-1 knockdown group, GSH levels were significantly decreased (Fig. 4J), indicating an enhancement of ferroptosis activity in KIRC cells.

3.5. Upregulation of DJ-1 alleviated erastin-induced ferroptosis in KIRC cells

Next, we explored whether DJ-1 overexpression prevented KIRC from erastin-induced ferroptosis. First, we activated the ferroptosis event in KIRC using erastin. Compared to the control, the overexpression group exhibited a notable decrease in mitochondrial morphological changes associated with ferroptosis, as shown by transmission electron microscopy (Fig. 5A). It was indicated that DJ-1 overexpression effectively inhibited erastin-induced ferroptosis in KIRC cells. Additionally, measurements of MDA levels (Fig. 5B), lipid ROS (Fig. 5C–F), and iron content (Fig. 5G) revealed a significant reduction in the overexpression group. In contrast, GSH levels were notably higher in the overexpression group than in the control group (Fig. 5H). Thus, these findings further underscored the critical regulatory role of DJ-1 in ferroptosis within KIRC cells.

3.6. DJ-1 acted on Hcy synthesis in the transsulfuration pathway to affect ferroptosis

The key pathway for combating ferroptosis is recognized as the Cystine/GSH/GPX4 axis, with critical proteins including the cystine/glutamate antiporter (System Xc^-) and GPX4 [13]. Through Western blot analysis, we evaluated the impact of DJ-1 silencing on these two crucial proteins. The results indicated that the downregulation of DJ-1 did not alter the expression levels of SLC7A11 and GPX4 (Fig. 6A). However, since the normal function of GPX4 depends on the presence of GSH, we further investigated the influence of DJ-1 on the upstream pathways of GPX4 [14]. First, ferroptosis was induced in the knockdown cell lines by utilizing erastin, together with pretreatment with exogenous GSH and N-acetylcysteine (NAC). The results showed that both lipid ROS levels and cell death were completely reversed by GSH and NAC (Fig. 6B, C). This result confirmed that DJ-1 regulates ferroptosis by impacting GSH synthesis. Next, we treated the erastin-induced knockdown cell lines with methionine (Met) and various intermediates of the transsulfuration pathway. The results revealed that only Hcy fully reversed the occurrence of ferroptosis in the knockdown cell lines (Fig. 6D and E). Using enzyme-linked immunosorbent assay (ELISA), Hcy levels were decreased in DJ-1 knockdown 786-O cells compared to the control group. Furthermore, after ferroptosis induction, Hcy levels decreased significantly (Fig. 6F), while no changes in SAH were noted (Supplementary Fig. S1A). Similarly, the overexpression group exhibited a significant increase in Hcy levels (Fig. 6G), but there was no difference in SAH (Supplementary Fig. S1B). After exogenous SAH supplementary, a significant increase in Hcy levels was observed in both the control and DJ-1 overexpression groups. Conversely, no change in Hcy levels was observed in the downregulated DJ-1 group (Fig. 6H and I). The experiments, approached from both positive and negative perspectives, consistently demonstrated that DJ-1 regulated SAH to Hcy synthesis in transsulfuration pathway to influence ferroptosis in KIRC.

3.7. The O-GlcNAcylation of DJ-1 at T19 residue stabilized DJ-1 homodimer and inhibited ferroptosis

Previous studies have indicated that DJ-1 could influence the interaction between SAHH (the only known enzyme which catalyzes the hydrolysis of SAH to Hcy) and its negative regulatory factor AHCYL1, but the exact mechanism remains unclear [15]. Our subsequent research focused on delving into this mechanism. O-linked β-N-

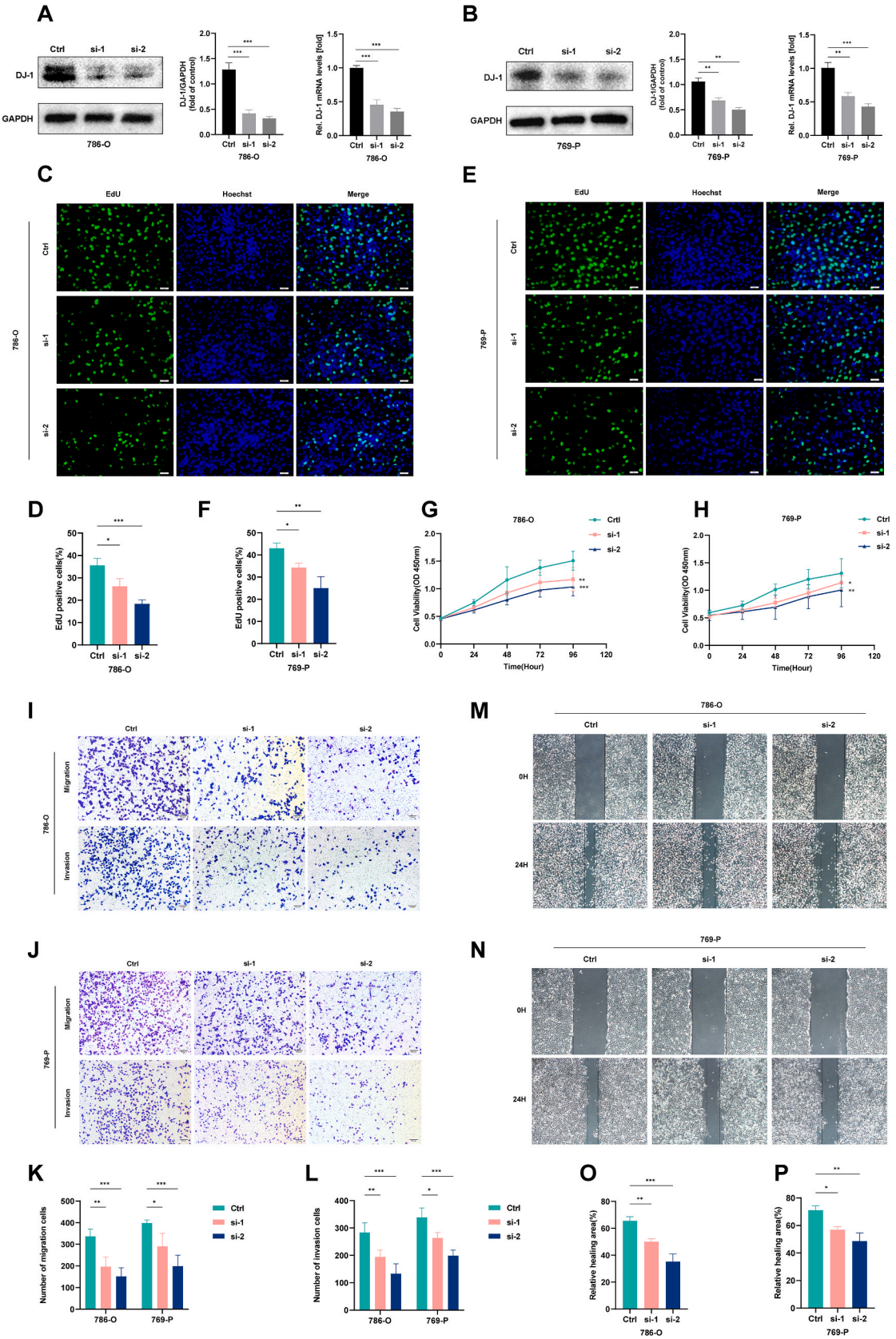


Fig. 2. Effects of DJ-1 downregulation on proliferation and metastatic ability in KIRC cells. A, B Evaluated the DJ-1 knockdown efficiency in KIRC cell lines by Western blotting and qRT-PCR. C–H Evaluation of cell proliferation activity following DJ-1 downregulation using EdU and CCK-8 assays. Scale bar: 50 μm. I-P Representative images and quantitative results of transwell and wound healing assays on DJ-1 knockdown KIRC cell lines. Scale bars: 100 μm and 200 μm. (All data are presented as mean ± standard deviation of three independent experiments. *P < 0.05, **P < 0.01, ***P < 0.001).

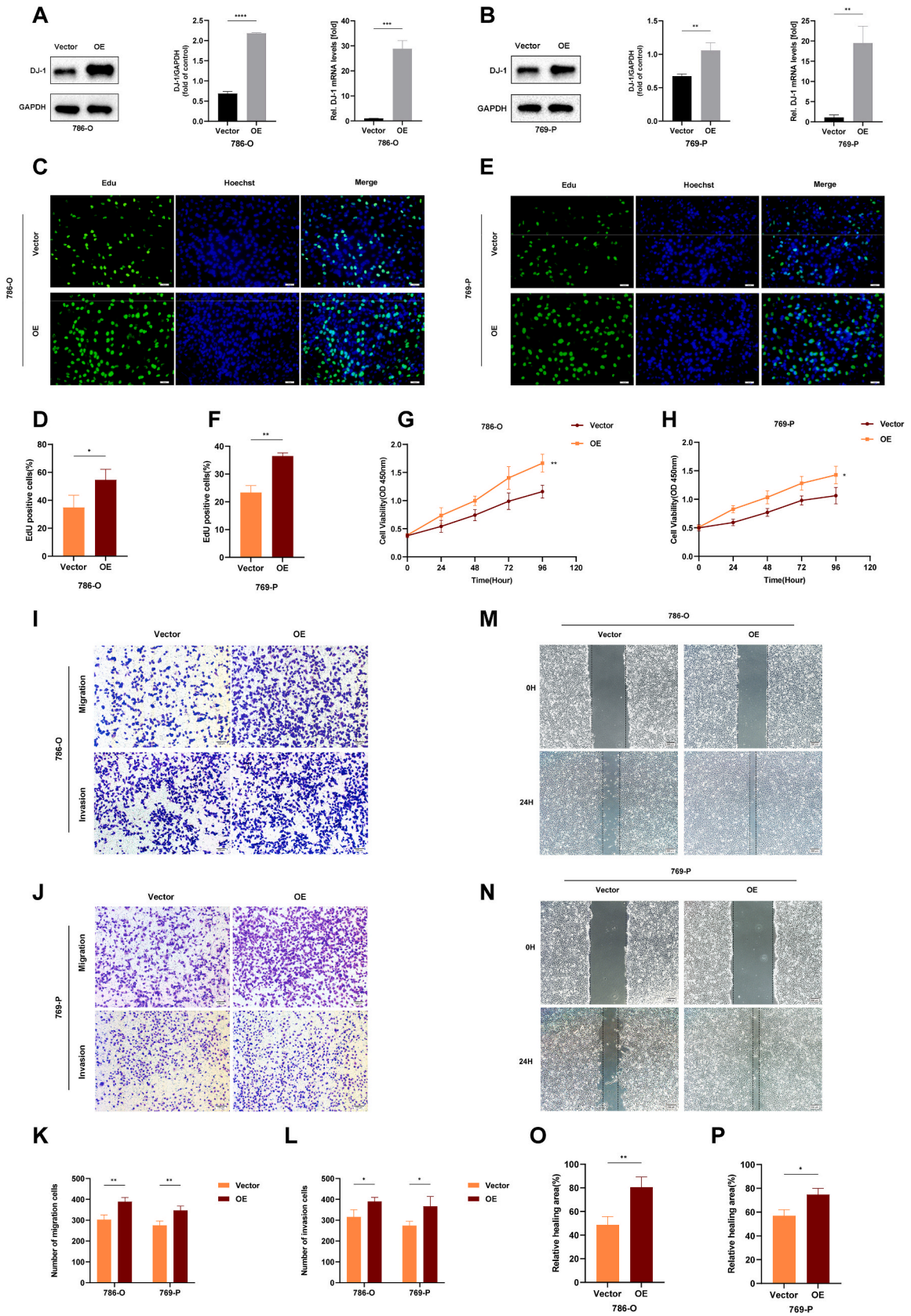


Fig. 3. Overexpression of DJ-1 enhances proliferation and metastatic ability in KIRC cells. A, B Established DJ-1 overexpression KIRC cell lines and evaluated by Western blotting and qRT-PCR. C–H Cell proliferation activity following DJ-1 overexpression was assessed through EdU and CCK-8 assays. Scale bar: 50 μm. I-P Representative images and quantitative results of transwell and wound healing assays on DJ-1 overexpression KIRC cell lines. Scale bars: 100 μm and 200 μm. (All data are presented as mean ± standard deviation of three independent experiments. *P < 0.05, **P < 0.01, ***P < 0.001, ****P < 0.0001).

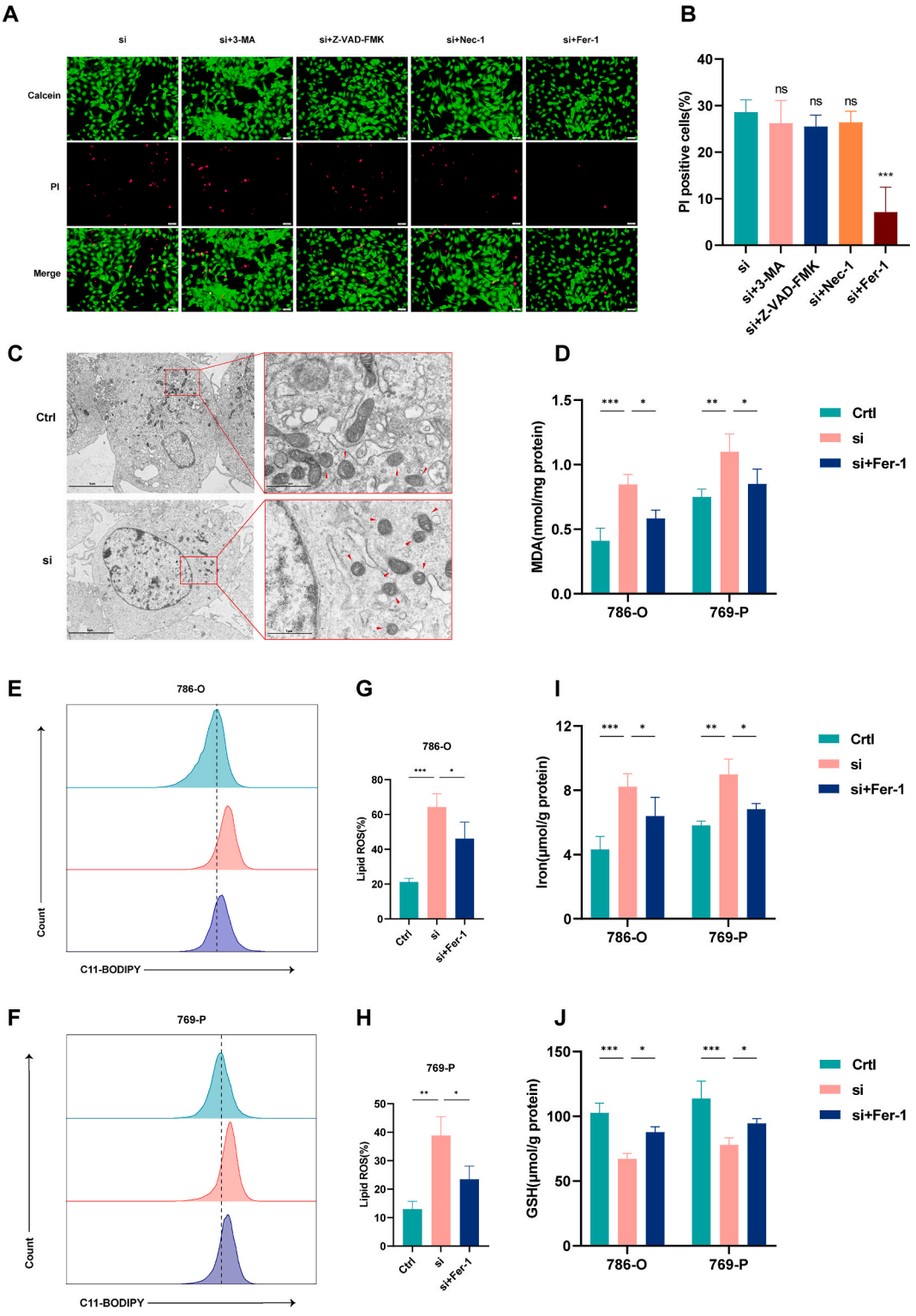


Fig. 4. Downregulation of DJ-1 induces ferroptosis in KIRC cells. A, B Treated DJ-1 knockdown 786-O cells with 3-MA (2 mM), Z-VAD-FMK (20 μM), Nec-1 (50 μM), and Fer-1 (1 μM) for 12 h, and performed cell viability assessment and statistical analysis. Scale bar: 50 μm. C Mitochondrial morphology in 786-O cells from the control and knockdown groups was examined using transmission electron microscopy, showcasing the most representative images. The red arrows indicated atrophied mitochondria. Comparison of levels of MDA (D), lipid ROS (E-H), iron content (I), and GSH content (J) was made among the control group, knockdown group, and knockdown + Fer-1 (1 μM) group in both 786-O and 769-P cells. (All data are presented as mean ± standard deviation of three independent experiments. *P < 0.05, **P < 0.01, ***P < 0.001). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

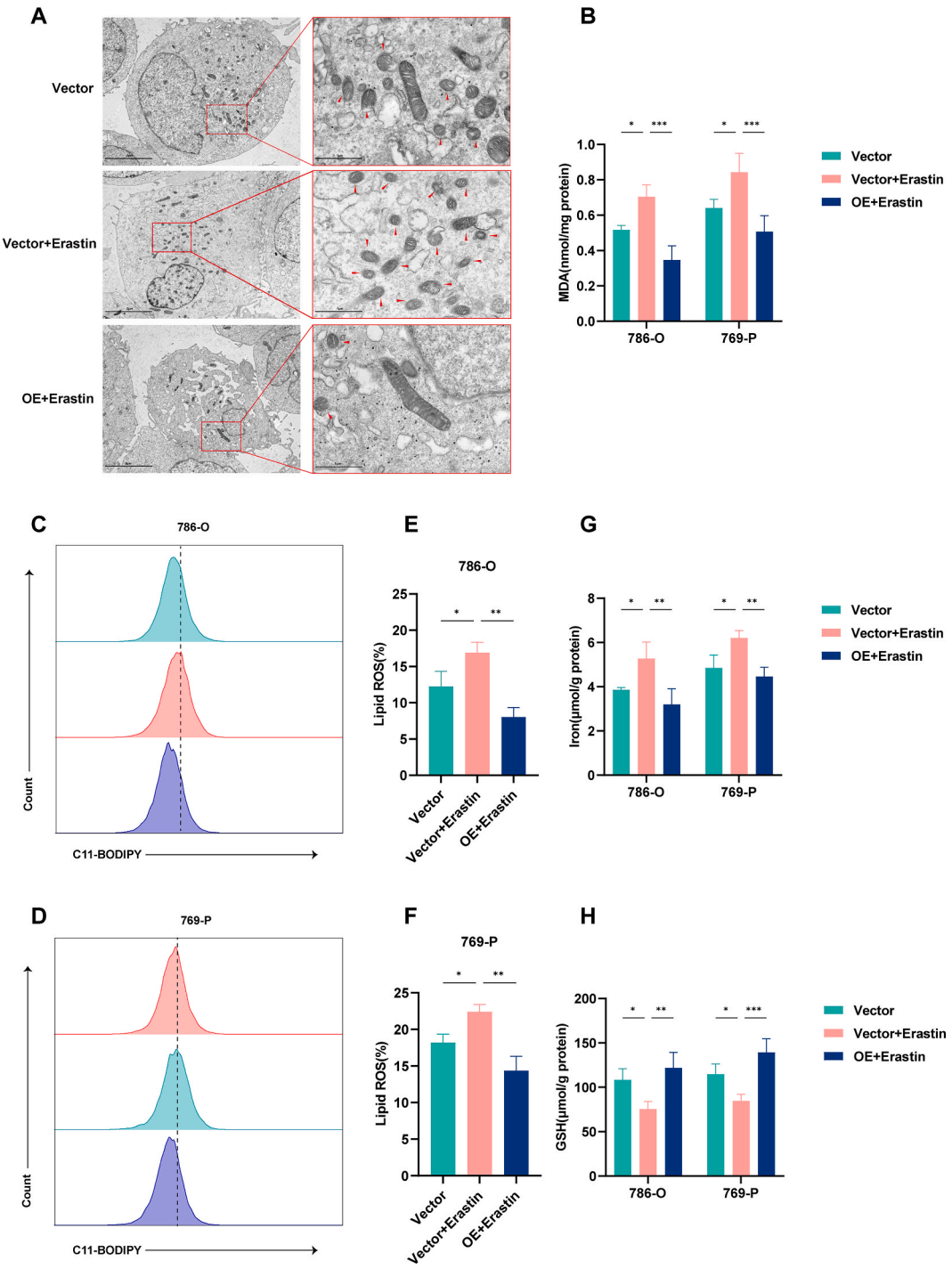


Fig. 5. Overexpression of DJ-1 attenuates Erastin-induced in KIRC cells. A Analyzed the mitochondrial morphological changes in 786-O cells from the Vector, Vector + Erastin (2 μM), and overexpression + Erastin (2 μM) groups using transmission electron microscopy, and the most representative images were selected. The levels of MDA (B), lipid ROS (C-F), iron content (G), and GSH content (H) were compared among the Vector, Vector + Erastin (2 μM) and overexpression + Erastin (2 μM) groups in both 786-O and 769-P cells. (All data are presented as mean ± standard deviation of three independent experiments. *P < 0.05, **P < 0.01, ***P < 0.001).

acetylglucosamine (O-GlcNAc) is a common form of glycosylation, which has been reported to play important roles in various cancers [16]. It was veiled that O-GlcNAc had impacts on regulation of protein functions, modulation of protein–protein interactions participation in cellular stress responses and so on [17]. It had been reported that DJ-1 possessed O-GlcNAcylation [18], and GSEA analysis results also indicated that DJ-1 was associated with diseases related to the biosynthesis of glycosylation precursors(Supplementary Fig. S1C). Thus, we sought to explore whether O-GlcNAcylation of DJ-1 participated in the regulation of ferroptosis. Firstly, it was verified that DJ-1 indeed had O-GlcNAcylation modification (Fig. 7A). Meanwhile, OGT interacted with and O-GlcNAcylated DJ-1 (Fig. 7B and C). As is well known, DJ-1 exists ubiquitously in a form of homodimer, which is crucial for its function [19]. So, we assumed that the O-GlcNAcylation of DJ-1 may take part in the regulation of ferroptosis through stabilizing DJ-1 homodimer. We found that knockdown of OGT led to a significant reduction in the O-GlcNAcylation of DJ-1, and meanwhile, the interaction of exogenous Flag- and Myc-DJ-1 was also inhibited, which suggested that the O-

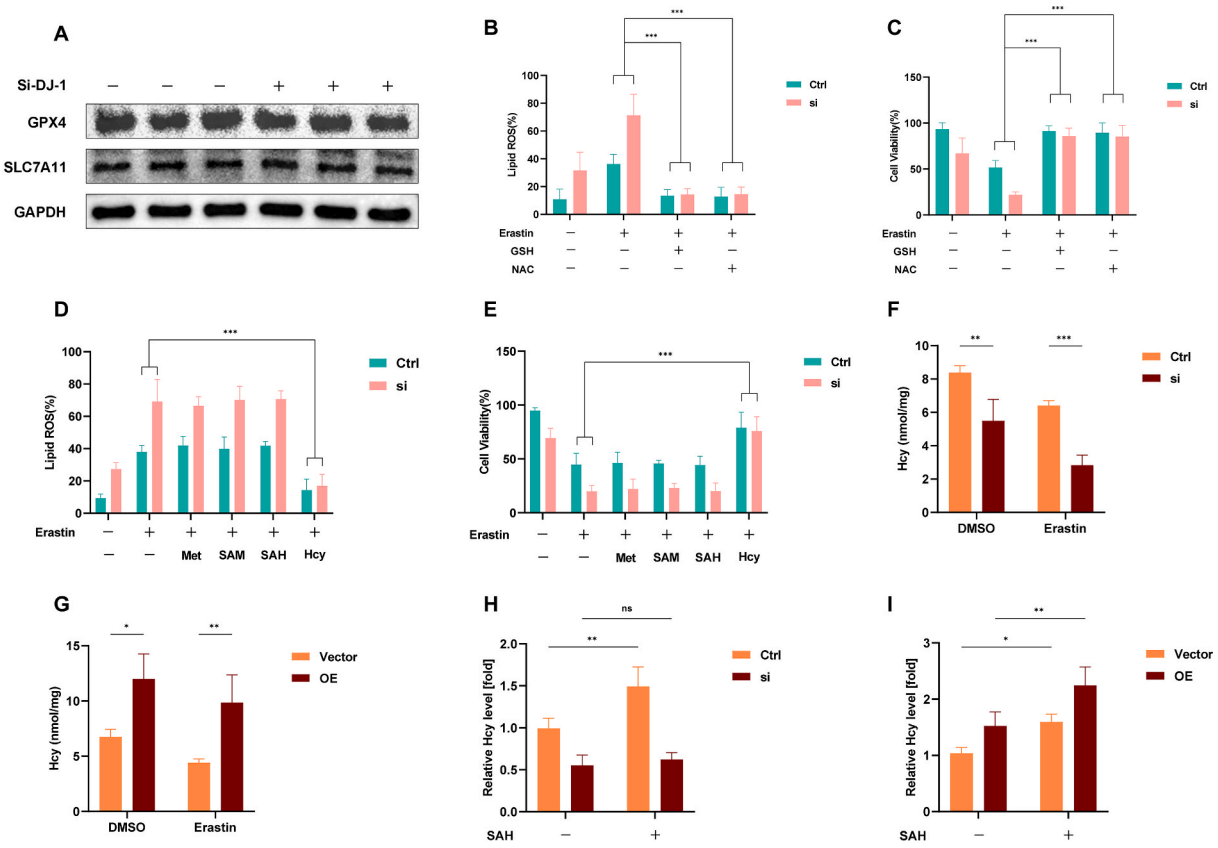


Fig. 6. DJ-1 regulates Hcy synthesis in the transsulfuration pathway to affect ferroptosis. A Used Western blotting to analyze the impact of DJ-1 knockdown on the expression of GPX4 and SLC7A11. B 786-O cells with downregulated DJ-1 were treated with Erastin (2 μM), with or without GSH (0.5 mM) and NAC (0.5 mM) for 12 h, after which lipid ROS generation was measured. C The viability of 786-O cells with downregulated DJ-1 was assessed following treatment with Erastin (2 μM), with or without GSH (0.5 mM) and NAC (0.5 mM) for 36 h. D Lipid ROS production was quantified in 786-O cells with downregulated DJ-1 that were treated with Erastin (2 μM), with or without indicated intermediates (Met, SAM, SAH, and Hcy. Each at 0.5 mM) for 12 h. E Cell viability was evaluated in 786-O cells with downregulated DJ-1 after treatment with Erastin (2 μM), with or without indicated intermediates (Met, SAM, SAH, and Hcy. Each at 0.5 mM) for 36 h. F, G The levels of Hcy in 786-O cells with downregulated and overexpressed DJ-1 were measured after 12 h of treatment with Erastin (2 μM). H, I Following exogenous addition of SAH (0.5 mM) for 12 h, Hcy levels were assessed in 786-O cells with downregulated and overexpressed DJ-1. (All data are presented as mean ± standard deviation of three independent experiments. *P < 0.05, **P < 0.01, ***P < 0.001).

GlcNAcylation of DJ-1 stabilized the homodimeric structure (Fig. 7C and D). Then, we verified that O-GlcNAcylation-deficient mutation of T19 residue, predicted by the GlyGen Portal (https://www.glygen.org) [20], exhibited markedly lower level of O-GlcNAcylation (Fig. 7E). Furthermore, it was showed that O-GlcNAcylation-deficient mutation T19A remarkably promoted the binding of SAHH and AHCYL1 (Fig. 7F). Subsequently, we found that deficiency of O-GlcNAcylation facilitated the ferroptosis-like cellular damage (Fig. 7G). Hence, O-GlcNAcylation of DJ-1 attenuated ferroptosis through stabilizing the homodimeric structure, which prevented the binding of SAHH and AHCYL1.

3.8. DJ-1 influenced the growth and progression of KIRC in vivo

To assess the influence of DJ-1 on KIRC progression in vivo, 786-O cells with either DJ-1 overexpression or knockdown were implanted into the subcutaneous area of the axilla in 6-week-old nude mice. The final results revealed that the knockdown group had slower tumor growth compared to the control group, showing significantly reduced tumor volume and weight. In contrast, tumors in the overexpression group exhibited a markedly accelerated growth rate, with both tumor volume and weight significantly exceeding those observed in the other groups (Fig. 8A–C). Furthermore, we performed immunohistochemical analysis on the tumors obtained from the mice, comparing the expression levels of DJ-1 and Ki67 among the different groups. Ki67 is a nuclear protein closely associated with the cell division cycle and is widely studied as an indicator of tumor cell proliferation capability [21]. In the overexpression group, the expression of DJ-1 and Ki67 was significantly increased, while a decrease in DJ-1 and Ki67 expressions were observed in the knockdown group (Fig. 8D–F). These findings imply that DJ-1 may also regulate the growth and metastasis of KIRC in vivo.

4. Discussion

Kidney cancer accounts for approximately 400,000 new cases each year, with a global mortality rate approaching 175,000, of which 75% of new cases are classified as KIRC [22]. Predictions suggest that kidney cancer incidence may rise over the next decade, underscoring the urgent need to address this pressing global health concern [23]. Recent advances in the understanding of ferroptosis have provided new prospects for cancer therapy. However, its specific role in KIRC therapy remains unclear, necessitating further research to elucidate its potential value. By utilizing data from the TCGA and GEO databases and performing immunohistochemical analysis, this study revealed that DJ-1 was significantly overexpressed in KIRC and correlated with a worse prognosis. Functional experiments revealed that downregulating DJ-1 expression inhibited the growth and metastasis of KIRC. Furthermore, our findings indicated that DJ-1 influenced the malignant phenotype of KIRC through the negative regulation of ferroptosis. Validation experiments via DJ-1 overexpression yielded consistent results. Mechanistically, O-GlcNAcylation of DJ-1 at T19 residue affected the interaction

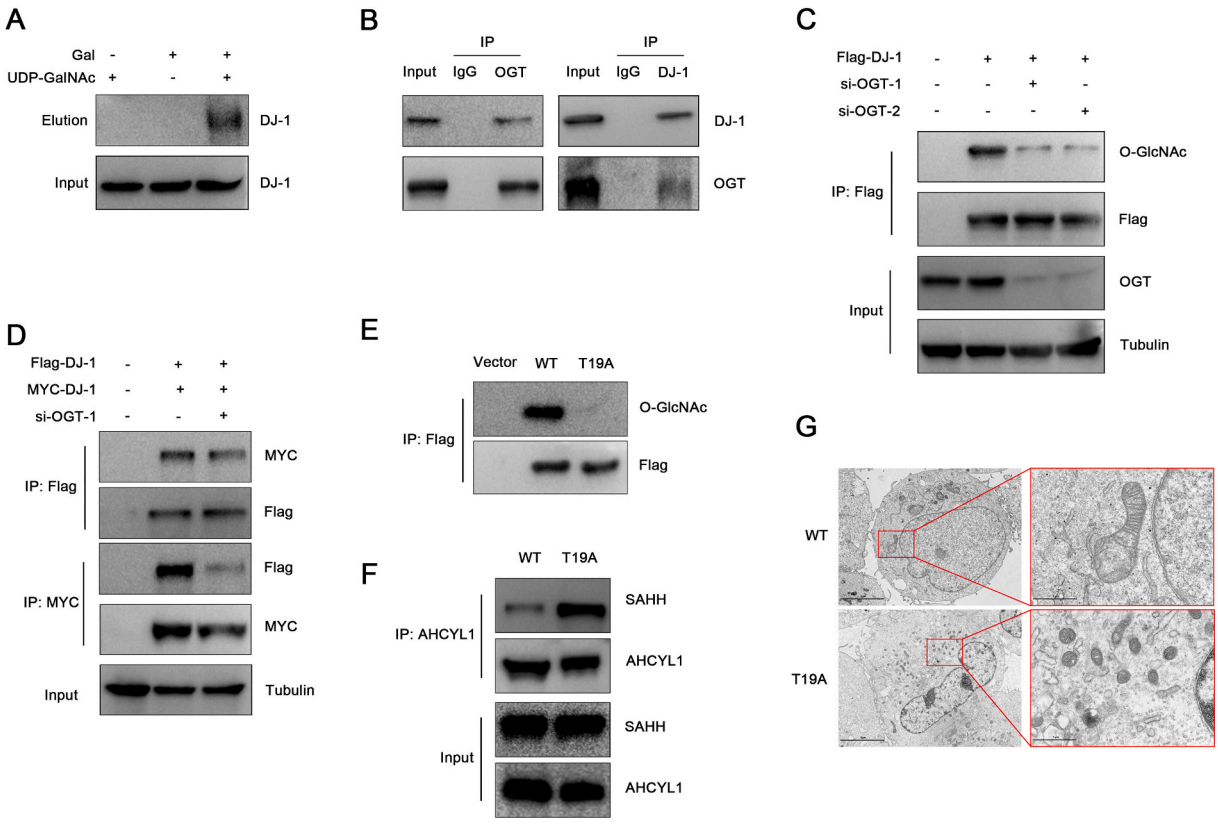


Fig. 7. OGT-mediated O-GlcNAcylation of DJ-1 at the T19 residue stabilized its homodimeric structure and affects ferroptosis by preventing the binding of SAHH and AHCYL1. A DJ-1 O-GlcNAcylation analysis using a chemoenzymatic labeling method. Immunoblot of O-GlcNAcylated DJ-1 in elution and total DJ-1 in input. B Reciprocal endogenous immunoprecipitation (IP) assay between OGT and DJ-1 in 786-O cells. C Transfected the indicated plasmids and si-OGT into 786-O cells, and performed IP using anti-FLAG beads and indicated antibodies. D The effect of OGT knockdown on DJ-1 homodimeric structure in 786-O cells was examined by CO-IP. E Western blotting results revealed the levels of O-GlcNAcylation of DJ-1-T19A and WT in 786-O cells. F Western blotting indicated alterations in the binding of SAHH and AHCYL1 after transfection of DJ-1- wt and DJ-1-T19A into 786-O cells. G Assess ferroptosis events in 786-O cells transfected with the indicated plasmids. (n = 3).

between SAHH and its negative regulator AHCYL1 within the transsulfuration pathway, thereby promoting GSH synthesis to regulate ferroptosis. Therefore, our study provides new insights into ferroptosis-based treatments and the molecular mechanisms underlying KIRC.

The central mechanism of ferroptosis involves regulating the equilibrium between oxidative and antioxidative systems [24]. Accumulation of iron ions triggers the Fenton reaction, generating a significant number of highly reactive hydroxyl radicals, which in turn lead to the accumulation of lipid ROS [25]. System Xc-, composed primarily of the light chain subunit (SLC7A11) and the heavy chain subunit (SLC3A2), functions as an amino acid reverse transporter. The primary functional component, SLC7A11, facilitates the exchange of extracellular cystine for intracellular glutamate [26]. Cystine is a critical rate-limiting substrate for GSH synthesis, while GPX4, a key enzyme in combating lipid peroxidation, relies on the presence of GSH as a cofactor for its normal function [27]. Thus, the SLC7A11/GSH/GPX4 axis is regarded as the most vital protective pathway against ferroptosis, and the inhibition of this pathway is closely linked to its onset. In our study, we observed an increased incidence of ferroptosis events in KIRC cells with downregulated DJ-1, accompanied by a significant decrease in GSH levels. Unexpectedly, no changes in the expression of SLC7A11 and GPX4 were detected in the downregulation group. Nevertheless, considering the essential role of GSH in GPX4 function, we speculated that DJ-1 might not directly influence the SLC7A11/GSH/GPX4 axis, but might affect GPX4 through upstream mechanisms.

The transsulfuration pathway functions as a compensatory mechanism for synthesizing GSH from Hcy, which is derived from serine and methionine during extracellular cysteine deficiency [28]. The pathway functions as a rapid adaptive response to environmental stimuli and has attracted considerable attention for its protective role against tumors [29]. Research demonstrates that HSF1 regulates cystathionine-β-synthase levels, affecting the conversion of Hcy to cystathionine and thus promoting a malignant phenotype in prostate cancer cells [30]. Additionally, MYCN enhances both cysteine uptake and the activity of the transsulfuration pathway in neuroblastoma, thereby increasing resistance to ferroptosis [31]. Knocking down CARS in various tumors has been shown to upregulate genes associated with the transsulfuration pathway, promote cysteine synthesis, and elevate intracellular GSH levels, consequently inhibiting the occurrence of ferroptosis [32]. These previous studies underscore the protective role of the transsulfuration pathway against ferroptosis in tumors. While this tumor-protective characteristic may demonstrate high background specificity, it continues to provide valuable theoretical support for follow-up research [33]. Through rescue experiments and ferroptosis detection following treatment with specific intermediates, we have preliminarily established that DJ-1 primarily influences the synthesis of Hcy in the transsulfuration pathway to exert its ferroptosis-resistant effects, revealing potential mechanisms for further investigation.

DJ-1, an essential antioxidant, primarily in the form of a homodimer characterized by a helix-loop-helix sandwich structure configuration [34]. Under mild oxidative stress, the Cys106 residue in DJ-1 is oxidized to its sulfinic form (Cys106-SO2H), while the overall structure of DJ-1 remains intact, resulting in its maximal antioxidant capacity. However, under more severe oxidative stress, the Cys106 residue can be further oxidized to its sulfonic form (Cys106-SO3H), resulting in instability of the homodimer state and consequent loss of DJ-1's activity [9]. Thus,

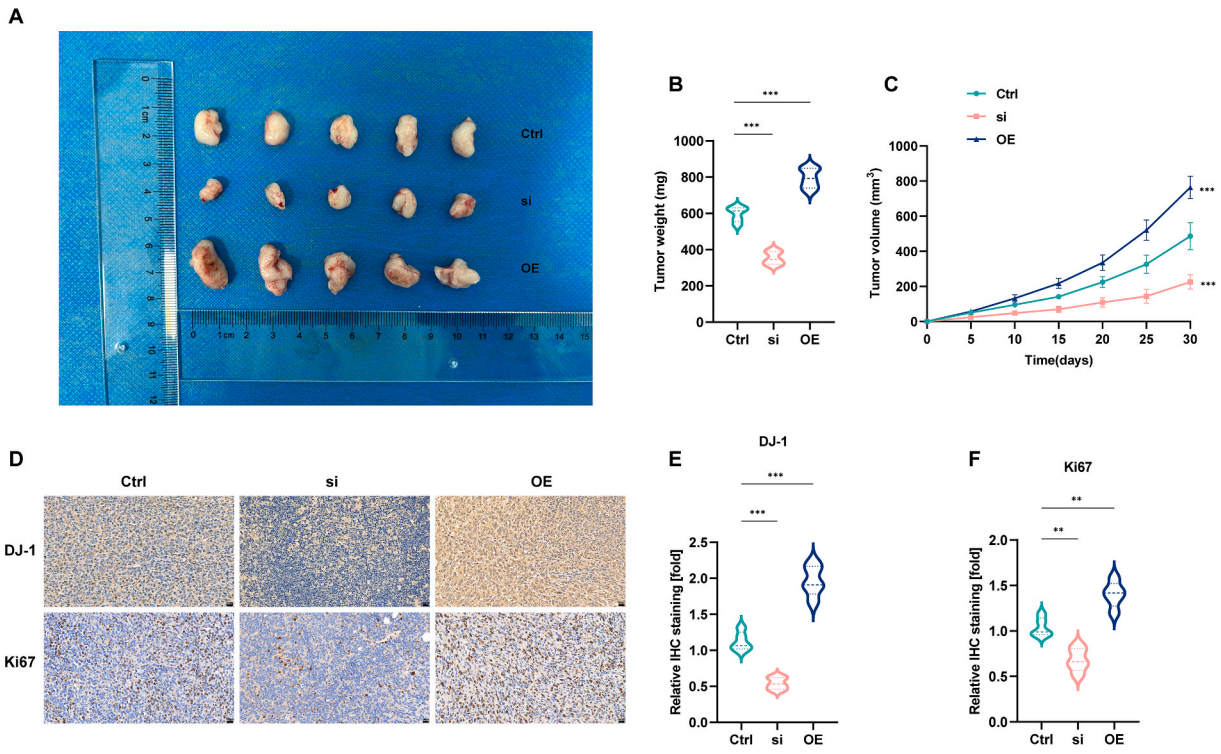


Fig. 8. DJ-1 influenced the growth and progression of KIRC in vivo. A Comparison of tumor sizes among different groups of nude mice. B Comparison of tumor weights among different groups of nude mice. C Measurement of tumor volume at specific time points after tumor implantation in nude mice. D-F Expression levels of DJ-1 and Ki67 in tumor tissues from nude mice, along with statistical analysis. Scale bar: 20 μm. (All data are presented as mean ± standard deviation of three and more independent experiments. *P < 0.05, **P < 0.01, ***P < 0.001).

the stability of the homodimeric structure is essential for DJ-1 to exert its antioxidant function. In subsequent mechanistic studies, we found that DJ-1 underwent OGT mediated O-GlcNAcylation, a post-translational modification that stabilized its homodimeric structure. The Co-IP results further revealed that blocking O-GlcNAcylation of DJ-1 at T19 residue prevented the binding of SAHH to the negative regulatory factor AHCYL1, thereby maintaining the tetrameric structure and activity of SAHH. The enzymatic activity of SAHH is essential for Hcy production, GSH synthesis, and resistance to erastin-induced ferroptosis [35]. Our research indicated that the O-GlcNAcylation of DJ-1 effectively mitigated the negative effects of AHCYL1 on SAHH, promoting the Hcy synthesis in the transsulfuration pathway to counteract ferroptosis. This finding provided a plausible explanation for the observed changes in GSH levels, despite no alterations in SLC7A11 and GPX4 in previous studies. O-GlcNAcylation, which modifies serine and threonine residues, plays a role in key cellular pathways by affecting transcription, translation, and protein balance. Its occurrence in tumor cells is strongly linked to cell death [36]. Specifically, inhibiting O-GlcNAcylation can enhance ferritin autophagy and mitochondrial autophagy, leading to the accumulation of unstable iron in tumor cells and inducing ferroptosis [37]. In liver cancer cells, O-GlcNAcylation of c-Jun stabilizes protein expression and transcriptional activity, regulating GSH synthesis to impede ferroptosis [38]. A recent study demonstrated that targeting USP8 leads to the loss of OGT activity, inhibiting SLC7A11 O-GlcNAcylation and promoting ferroptosis in hepatocellular carcinoma [39]. It is worth noting that, while our experiments revealed potential mechanisms by which DJ-1 inhibits ferroptosis, it is likely that DJ-1 possesses additional, yet-to-be-explored functions. Nrf2 has garnered significant attention due to its remarkable antioxidant properties, and studies suggest that DJ-1 acts as an upstream regulator of Nrf2 [40,41]. The activation of DJ-1 enhances Nrf2 activity, thereby increasing the cell's antioxidant capacity. Moreover, the activation of Nrf2 may be a key factor conferring resistance to ferroptosis through the transsulfuration pathway [42]. Consequently, further investigation into the interaction between DJ-1 and Nrf2, as well as their roles in ferroptosis, is essential.

In summary, our study elucidated the gene expression model of DJ-1 in KIRC and its association with worse prognosis. The expression of DJ-1 was closely related to ferroptosis events and malignant phenotypes in KIRC, and it could promote tumor growth in vivo. Importantly, rescue experiments and metabolic analyses pointed out that DJ-1 inhibits ferroptosis by acting on the synthesis of Hcy in the transsulfuration pathway. Furthermore, additional experiments revealed that OGT mediated O-GlcNAcylation of DJ-1 at T19 residue stabilized its homodimeric structure, and this post-translational modification inhibited the binding of SAHH to the negative regulator AHCYL1, thereby restoring the activity of SAHH. Therefore, the final conclusion drawn was that O-GlcNAcylation of DJ-1 maintained the antioxidant function of the transsulfuration pathway, thereby reducing the occurrence of ferroptosis events in KIRC.

Ethical approval.

Animals used in the experiments were maintained in the specific pathogen-free animal center. All experimental protocols and animal care procedures received approval from the Ethics Committee at Renmin Hospital of Wuhan University and were carried out according to the committee's guidelines (WDRM 20240405B).

Author contributions

ZY, YL, XL, ZC, and HL designed the experiments. ZY, YL, WL and KX conducted the experiments, MW and YL carried out the animal studies. ZY and MW handled the processing and analysis of the data. ZY and YL wrote the manuscript. HL, XL, WL and ZC guided the manuscript revisions and ensured funding. All authors contributed to the study and approved the final manuscript.

CRediT authorship contribution statement

Zhiwei Yan: Writing – original draft. **Yanze Li:** Writing – original draft, Visualization. **Minghui Wang:** Writing – original draft. **Kai Xu:** Supervision, Software. **Yunxun Liu:** Visualization, Validation. **Lei Wang:** Writing – review & editing. **Hongbo Luo:** Writing – review & editing. **Zhiyuan Chen:** Writing – review & editing. **Xiuheng Liu:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.intimp.2025.114098.

Data availability

Data will be made available on request.

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